

CSCE 636 DEEP LEARNING · MODULE 13 · LECTURE NOTES

Shipping a Model

Inference cost, and what you can honestly claim.

MODULE OUTCOMES

- Reason about the inference cost budget.
- Apply quantisation and measure the quality cost.
- Apply distillation and pruning.
- Explain serving considerations — batching, caching, latency.
- State what a trained model can honestly promise.

1 The inference cost budget

INFERENCE DOMINATES THE LIFETIME COST

Training happens once, or periodically on a retraining schedule. Inference happens on every single request, forever. For any model with real traffic the integral over inference exceeds the training cost within weeks.

So a model that costs twice as much to train and half as much to serve is usually the better choice — and the research literature optimises the wrong one of these, reporting training compute and benchmark accuracy while inference cost appears, if at all, as a footnote.

And inference is frequently memory-bandwidth-bound rather than compute-bound. At batch size one, every weight is read from memory and used in exactly one multiply-accumulate — **arithmetic intensity is close to its worst possible value**, and the accelerator's arithmetic units are mostly idle (CSCE 735 Module 07's roofline, at the far left).

Which explains why quantisation helps as much as it does: halving the bits per weight halves the bytes that must be moved, and **the speedup comes from bandwidth rather than from faster arithmetic** — which is why int8 inference can be close to 4× faster even on hardware whose int8 arithmetic throughput is not 4× its fp32 throughput.

2 Making it smaller

Technique	What it does	Typical result
Post-training quantisation	Cast the trained weights to int8 or fp8, with per-channel scale factors calibrated on a small sample.	About 4× smaller, a small accuracy loss, and no retraining at all. The first thing to try.
Quantisation-aware training	Simulate the quantisation during training so the model adapts to it.	Noticeably better quality at low bit-widths, at the cost of a retraining run.
Distillation	Train a small student model to match a large teacher's output distribution.	The student frequently beats the same architecture trained directly on the labels — see below.
Structured pruning	Remove entire channels, attention heads (Module 07 §2), or layers.	A real wall-clock speedup, because the resulting dense model is simply smaller and the hardware is unchanged.
Unstructured pruning	Zero out individual weights wherever they are small.	Very high sparsity is achievable and rarely translates into wall-clock gains — a sparse matrix multiply costs more per nonzero than a dense one, so 90% sparsity can be slower than dense without specific hardware support. A theoretical speedup is not an achieved one.

DISTILLATION TRANSFERS MORE THAN THE LABELS

Train a small model to match a large model's full output *distribution* — typically with a softened softmax — rather than only its top prediction.

The soft targets carry information the hard label does not. That an image is rated 70% cat, 25% lynx, and 5% dog tells the student something about the similarity structure of the classes: that cats and lynxes are confusable in a way cats and dogs are not. A one-hot label contains none of that.

So the student frequently outperforms the identical architecture trained directly on the hard labels with the same data — which is counterintuitive, since the teacher's outputs are strictly less accurate than the ground truth.

Hinton called it 'dark knowledge' and the effect is robust across domains. It also means a distilled model inherits the teacher's biases and shortcuts (Module 12 §2) — the student is learning to imitate the teacher, including where the teacher is wrong, and evaluating the student only against the teacher will never reveal this.

3 Serving

- **Batching is the single largest lever.** It amortises each weight read across many inputs, **converting a memory-bound problem into a compute-bound one** (§1) — and it **adds latency**, because requests wait for the batch to fill. **The same latency-for-throughput trade as CSCE 678 Module 02's buffers and CSCE 650's frame pacing.**
- **Continuous (in-flight) batching for generation.** Admit new requests into the batch as earlier ones finish, rather than waiting for every sequence in a batch to complete — a large throughput gain for variable-length generation.
- **KV caching.** Reuse the attention keys and values computed for previous tokens rather than recomputing them each step. **Essential — without it generation is quadratic — and the cache becomes the memory bottleneck**, growing with batch size times sequence length, which is what PagedAttention and similar schemes manage.
- **Compilation.** Kernel fusion, graph-level optimisation, and target-specific code generation — **which is CSCE 605's subject applied here**, and the same argument as that course's Module 13 made about shader compilation, including the warm-up cost.
- **And measure the p99 rather than the mean** (CSCE 678 Module 13 §2) — a serving system's tail is what users experience, and batching makes the tail worse in exchange for the throughput.

4 Claiming honestly

- **'Top-1 accuracy 82.3% on this benchmark, against a from-scratch baseline of 71.1%, split by subject, data collected in 2024.'** The baseline, the split key, and the date are what make it a claim (CSCE 633 Module 01 §4).
- **'Worst-performing slice: 64%, n = 290, on the under-represented device type.'** The slice named and counted (CSCE 633 Module 12 §3).
- **'Degrades to 61% under the standard corruption benchmark, and fails on inputs outside the training distribution without signalling that it has done so.'** Stating the known failure mode is more useful than another point of accuracy (Module 12).
- **'int8 quantised: 3.8× smaller, 2.1× faster at batch size 1, costing 0.6 points of top-1.'** The compression trade, measured in all three dimensions.
- **And what you cannot honestly say: 'human-level', 'understands', or 'robust'. 'Human-level' requires stating which humans, on which task, under which conditions, and with what inter-annotator agreement** (CSCE 633 Module 11 §4) — usually it means 'exceeds the accuracy of the annotators who produced the labels', which is a statement about the labels. **Every course in this program ends on this point, and the repetition is the argument.**

WHERE THIS LEAVES YOU

You can build a network from automatic differentiation upward, diagnose why it will not train, choose an architecture from what the data's structure actually is, scale it when it does not fit on one device, and compress it when it must be served.

And you know what these models fail at — adversarially, through shortcuts learned from the collection process, and confidently off-distribution — **none of which the benchmark number tells you**, and all of which decide whether a deployment works.

CSCE 633 supplied the evaluation discipline this course inherited — baselines, slices, leakage, honest claims. **CSCE 669 supplies the optimisation theory underneath Module 03**, which is where the semester finishes. **And Modules 10 and 11 closed the loop back to CSCE 647's denoising and CSCE 748's reconstruction**, which is why this course sits in a graphics program at all.

The discipline does not change. Baseline first, measure what you claim, report by slice, and never say more than you established. That has now been the closing paragraph of eight consecutive courses, and it is the part of this degree that will still be true when every architecture in it is obsolete.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Hinton, Vinyals & Dean — Distilling the Knowledge in a Neural Network (free)	The §2 'dark knowledge' argument , from the paper that named it.
Jacob et al. — Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference (free)	The §2 quantisation scheme, including quantisation-aware training.
Kwon et al. — Efficient Memory Management for Large Language Model Serving (PagedAttention / vLLM) (free)	The §3 KV cache problem , and a solution borrowed directly from virtual memory (CSCE 611).
Blalock et al. — What is the State of Neural Network Pruning? (free)	The §2 caution — a survey finding that pruning results are frequently not comparable and that reported speedups are often theoretical.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Measure your model's inference latency and memory** at batch sizes 1, 8, and 64.

- 2 **Place inference on a roofline** and confirm it is memory-bound at batch 1.
- 3 Apply post-training int8 quantisation and measure size, speed, and accuracy.
- 4 **Apply quantisation-aware training** and compare at 8 and 4 bits.
- 5 **Distil your model into one a quarter the size** and compare against training that architecture directly.
- 6 **Check whether the student inherited a shortcut** from the teacher (Module 12).
- 7 Apply structured and unstructured pruning at equal sparsity and **measure the actual wall-clock speedup of each**.
- 8 Implement KV caching and measure the memory growth with sequence length.
- 9 **Measure p50 and p99 latency under load**, with and without batching.
- 10 **Project 2 is now due**. Submit the model, the training curves including failures, the baseline comparison, the cost measurements, the failure probe, and the honest claim.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why does inference dominate lifetime cost, and what does the literature optimise?
- 2 Why is small-batch inference memory-bound, and what follows for quantisation?
- 3 Compare five compression techniques and their real results.
- 4 Why does unstructured sparsity rarely give a wall-clock speedup?
- 5 Why does distillation beat direct training, and what does the student inherit?
- 6 Name five serving considerations and the trade batching makes.
- 7 Give four honest claims and three things you cannot say.